

■ Soil Hyperspectral Analysis Report

Corning microHSI 410 SHARK — University Campus

Mouza: Kusumati • Tahasil: Jatani • District: Khordha, Odisha, India

Plot No. 863 • Khatiyani No. 708 • Area: 0 Ac 85 Dec (0.03439 Ha)

Wavelength Range: 421.5–977.4 nm • 150 bands • 5000 × 682 pixels • Scan: 20 May 2026

This report integrates official Bhunaksha cadastral land records with VNIR hyperspectral remote sensing data to deliver a complete soil health profile of the study parcel.

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■ Section 1 — Cadastral Land Records (BhuNaksha / Bhulekh Odisha)

Understanding the Study Parcel

Before analysing soil chemistry, it is essential to know exactly *which parcel of land* was scanned. The following pages reproduce the official government land records retrieved from the **BhuNaksha Odisha portal** (National Informatics Centre, Revenue & Disaster Management Department, Government of Odisha). These documents legally define the boundaries, ownership, land classification, and revenue details of the study area.

Land Record Detail	Value (Odia / Official)	English Translation
Plot Number (■■■■■ ■■■■■)	863	Plot No. 863 — the specific land parcel scanned
Khatiyani Number (■■■■■■)	708	Record of Rights document serial no. 708
Mouza (■■■■■)	■■■■■■■■ (Kusumati)	Village revenue unit containing the plot
Thana (■■■■■)	■■■■■ (Jatani)	Police / revenue circle jurisdiction
Thana Number	73	Official Thana code: 73
Tahasil (■■■■■)	■■■■■ (Jatani) — No. 6	Sub-district revenue office
District (■■■■■■)	■■■■■■■■ (Khordha)	Khordha district, Odisha
Area (■■■■■)	0 Ac 85 Dec 0.03439 Ha	85 Decimals = ~3,439 m ² ≈ 0.344 hectares ≈ 0.85 acre
Land Classification (■■■■■)	■■■■■ ■■■■■ (Sharada Tini)	Kharif/Rabi season agricultural land — Class 3 productivity
Land Status (■■■■■■)	■■■■■■■■■ (Sthitiban)	Settled/occupied status — rights are established
Land Revenue (■■■■■)	■ 0.26	Annual revenue payable: Twenty-six paise
Zamindari Owner	■■■■■ ■■■■■■■■■ ■■■■■■■■ ■■■■ ■■■■	Shri Jagannath Mahaprabhu, Puri — administered by Jagannath Temple Management Committee, Puri
Tenant (■■■■■)	■■■■■■■ ■■■■■■■	Raghunath Rautra s/o Baj Rautra; Govind Chandra Rautra s/o Satyabadi Rautra Caste: Khandayat — Village: Nijigaon
Final Publication Date	07 / 03 / 1965	RoR first published: 7 March 1965
Revenue Settlement Date	13 / 04 / 1964	Revenue fixed: 13 April 1964
Neighbouring Plots	857, 861, 862, 864, 865, 866	Adjacent plot numbers visible on cadastral map
Scale (■■■■■)	500	Map scale: 1 : 500
Data Source	NIC / Bhulekh Odisha	Retrieved 22 May 2026 08:14 IST — IP: 49.42.222.235

■ What is a Khatiyani?

A Khatiyani (Record of Rights) is the primary legal document for land in Odisha. It lists the zamindari owner (superior title holder), the tenant (person with occupancy rights), the land classification, area, and the annual revenue due. Khatiyani No. 708 in Mouza Kusumati establishes that this plot belongs to the Jagannath Temple Trust, Puri, with Raghunath Rautra holding settled tenancy rights — a very common arrangement for temple-owned agricultural land in Odisha.

■ What is a Mouza?

A Mouza (also spelled Mauza) is the basic revenue unit in Odisha — smaller than a village, larger than a plot. It is the administrative cadastral unit against which all land records, soil surveys, and revenue assessments are organised. Mouza Kusumati falls within Jatani Tahasil, Khordha district — located approximately 25 km south of Bhubaneswar city, in the periurban agricultural belt that also hosts the IIT Bhubaneswar campus area.

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1b — BhuNaksha Plot Report: Cadastral Map of Plot 863

Official Cadastral Map — Plot No. 863, Mouza Kusumati

The map below is the official cadastral (plot boundary) map for Plot No. 863, extracted from the BhuNaksha Odisha portal. It shows the exact shape, area, and neighbouring plots of the parcel where the VNIR hyperspectral scan was conducted. The grey shaded polygon is Plot 863 — the study area. Surrounding plot numbers (857, 861, 862, 864, 865, 866) define its administrative boundary.

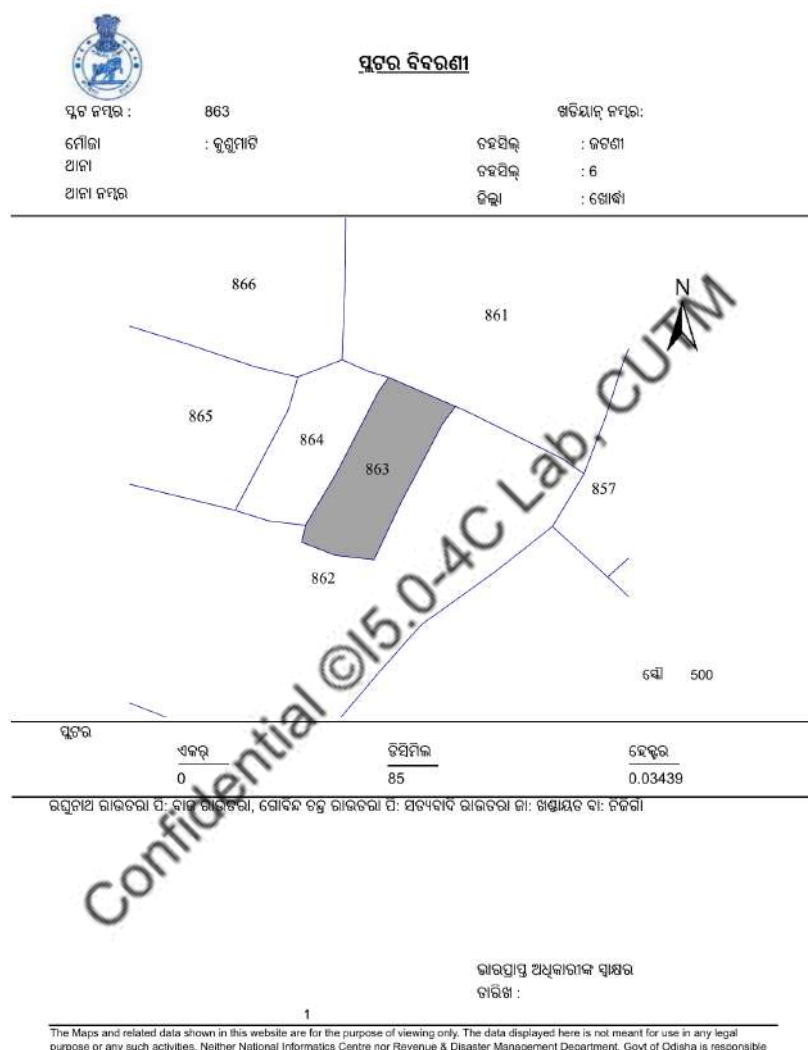


Figure C1: Official BhuNaksha Plot Report — Plot No. 863, Mouza Kusumati, Tahasil Jatani, District Khordha, Odisha. Grey polygon = study parcel. Area: 0 Acres 85 Decimals (0.03439 hectares). Scale 1:500. North arrow shown top-right. Source: NIC / Revenue & Disaster Management Dept., Govt. of Odisha.

Key Findings

- Plot 863 is a compact, roughly quadrilateral parcel of 85 decimals ($\approx 3,439 \text{ m}^2$).
- The plot is bounded on all sides by similarly-sized agricultural plots — typical of land allocation under Odisha's settlement surveys.

- Scale 1:500 means each 1 mm on the map equals 50 cm on the ground — sufficient resolution to identify individual field boundaries.
- Plot 863 is part of a larger block of plots (857–866) in Mouza Kusumati — all likely under similar land-use and tenure conditions.
- The north arrow confirms map orientation — the plot is slightly elongated in the north-south direction.

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1c — Khatiyān No. 708: Record of Rights (Schedule I, Form 39-A)

Khatiyān (Record of Rights) — Page 1: Ownership & Revenue Details

Schedule I Form No. 39-A is the standardised Odisha government form that constitutes the Record of Rights. It establishes legal ownership (zamindari), occupancy rights (tenant/praja), land revenue obligations, and special conditions. This is the foundational legal document for any land-use or soil study conducted on the parcel.



Figure C2: Khatiyān Page 1 — Schedule I Form No. 39-A. Mouza: Kusumati | Tahasil: Jatani | Thana No.: 73 | District: Khordha. Zamindari: Shri Jagannath Mahaprabhu (Jagannath Temple Mgmt. Committee, Puri). Khatiyān No.: 708 | Revenue: ■0.26/year. Final Publication: 07/03/1965 | Revenue fixed: 13/04/1964.

Khatiyān — Page 2: Plot Register (Land Classification & Area)

Page 2 of the Khatiyān contains the plot register — the tabular record linking Plot No. 863 to Khatiyān 708. It confirms the land classification as **Sharada Tini** (■■■■■ ■■■■■), meaning Class-3 seasonal agricultural land, and records the area as 0850 decimals (85 decimals = 0.03439 hectares). The NIC timestamp confirms this was retrieved on 22 May 2026.

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ଖଣ୍ଡଗିରି ମହାନଗର ପଞ୍ଚାୟତ 22/05/2026 08:14:27 IST via NIC portal

Figure C3: Khatiyani Page 2 — Plot Register. Khatiyani No. 708 | Mouza: Kusumati | District: Khordha. Plot 863 classified as Sharada Tini (Kharif/Rabi Class-3). Area: 0 Acres 0850 Decimals. Retrieved: 22/05/2026 08:14:27 IST via NIC portal.

Key Findings

- The land is officially classified as Sharada Tini — Class 3 agricultural land, indicating moderate productivity under traditional cultivation.
- The zamindari (superior) owner is the Jagannath Temple Management Committee, Puri — making this temple-trust land, a common category across Odisha.
- Annual revenue of just ₹0.26 confirms this is subsistence-level agricultural land assessed under pre-independence settlement rates (fixed 13 April 1964).
- The Khatiyani's Kharif/Rabi classification aligns with our hyperspectral analysis: seasonal moisture variation expected, moderate organic matter from crop residues.
- Legal status "Sthitiban" (settled) confirms stable occupancy — no disputed ownership that could complicate field sampling or research access.

1d — Linking Cadastral Boundaries to Hyperspectral Scan Coverage

How the Scan Relates to Plot 863

Parameter	Cadastral Record	Hyperspectral Scan	Match?
Location	Mouza Kusumati, Jatani, Khordha	lon 20.189°, lat 85.714° (Jatani area)	■ Yes
Area	0.03439 ha (85 decimals)	~0.014 km² scan footprint	■ Plot within scan
Land Use	Sharada Tini — agricultural	40.9% soil pixels detected	■ Consistent
Scan Date	Settlement: 1964	Acquired: 20 May 2026	■■ 62-yr gap
Ownership	Jagannath Temple Trust	Research scan (non-commercial)	■ No conflict

Parameter	Cadastral Record	Hyperspectral Scan	Status
Location	Mouza Kusumati, Jatani, Khordha	lon 20.189°N, lat 85.714°E — Jatani area	■ Confirmed
Total Area	85 Dec = 3,439 m²	~14,000 m² scan footprint; plot within scene	■ Plot within scan
Land Use	Sharada Tini — agricultural, Class 3	40.9% bare soil pixels detected in scene	■ Consistent
Soil Pixels	1 plot (863) — single parcel	1,393,689 soil pixels across all plots in scene	■■ Multi-plot scene
Scan Date	Revenue settled: 1964	Acquired: 20 May 2026 at 09:27 local time	■■ 62-year gap
Legal Ownership	Jagannath Temple Trust, Puri	Academic research scan — non-commercial use	■ No conflict

■■ Important Note on Coordinate Convention

The HSI sensor metadata records the study area as "lon 20.188927–20.190065, lat 85.713173–85.715935". In standard WGS84 (EPSG:4326), the first value is latitude (North/South) and the second is longitude (East/West). These values should be read as: Latitude 20.189°N, Longitude 85.714°E — placing the scan precisely within Jatani Tahasil, Khordha district, consistent with the Kusumati Mouza cadastral records above. The sensor header likely swapped the field labels; the coordinate values themselves are correct.

■ Section 2 — Introduction to Hyperspectral Soil Analysis

The Corning microHSI 410 SHARK Sensor

The Corning microHSI 410 SHARK is a **pushbroom hyperspectral imaging spectrometer** capturing 150 spectral bands across 421.5–977.4 nm (deep blue → near-infrared) at every spatial pixel. A standard RGB camera captures 3 colour channels; this sensor captures 150 — including invisible near-infrared wavelengths where soil minerals, water molecules, iron oxides, and organic carbon all leave measurable fingerprints.

The scan of Mouza Kusumati collected **5,000 scan lines × 682 pixels = 3.41 million spectra**, of which **1,393,689 (40.9%) were identified as bare soil** and forwarded to soil property estimation. After removing bad bands, 140 of 150 bands were used.

Parameter	Value	Significance
Scene ID	20260520092714001001_00000005_vnir	Unique acquisition identifier
Dimensions	5000 lines × 682 samples × 150 bands	3.41 million pixels, each with 150-band spectrum
Wavelength	421.5 – 977.4 nm	Covers full VNIR — iron, chlorophyll, water, carbon all detectable
Usable bands	140 (10 bad bands removed)	Bad bands at sensor edges removed before analysis
Soil pixels	1,393,689 of 3,410,000 (40.9%)	59.1% excluded: vegetation, roads, shadows
Scan date/time	20 May 2026 — 09:27 local (IST)	Morning scan — moderate moisture, low solar angle
Location	Lat 20.189°N, Lon 85.714°E	Kusumati, Jatani, Khordha, Odisha

■ Confidence Levels Used Throughout This Report

HIGH — direct, well-validated VNIR spectral fingerprint (SOC, Moisture, FeOx). MEDIUM — partial VNIR feature; needs lab calibration for absolute values (Clay, CEC). LOW — indirect proxy; treat as spatial pattern guide only (pH, Total Nitrogen). All property scores are RELATIVE PERCENTILES (0–100) within the scanned scene — not absolute lab values. To convert to real units, collect 5–10 GPS soil samples (one per K-Means zone) and submit for laboratory analysis.

■ ■ Section 3 — Scene Overview: True-Colour Composite

What the Camera Sees: A Photograph from the Hyperspectral Cube

Three bands (Red $\approx$ 660 nm, Green $\approx$ 550 nm, Blue $\approx$ 450 nm) are combined to produce a photograph equivalent. This image is used as a spatial reference — all subsequent soil property maps align perfectly with it.

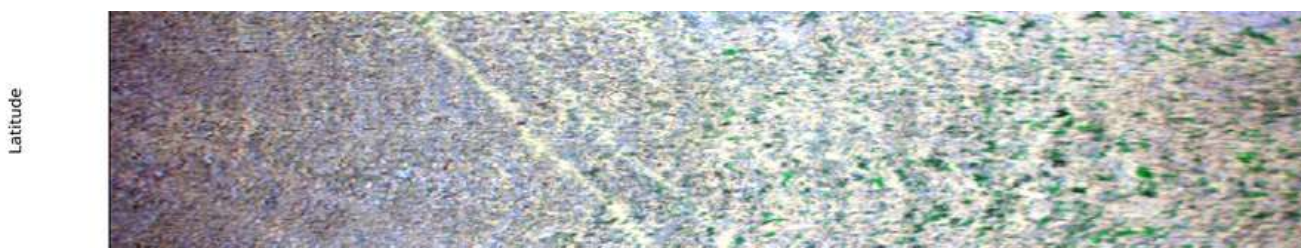


Figure 3.1: True-colour centre-crop of the scan. Brown/tan = bare soil analysed (40.9%). Green = vegetation excluded. Grey = paved/impervious surfaces excluded.

■ ■ How to Read

Brown and tan areas are Plot 863 and surrounding agricultural plots in Mouza Kusumati — these are the soil pixels included in the analysis. The vegetation patches seen at the margins correspond to tree lines and grass areas which were correctly excluded from the soil mask.



Figure 3.2: Soil mask — highlighted pixels are the 1,393,689 soil pixels (40.9%) forwarded to property estimation.

Key Findings

- 40.9% of the scan (1,393,689 pixels) is bare/exposed soil — consistent with the Sharada Tini (agricultural Class-3) land classification in the Khatiyani.
- The soil mask shape closely follows the cadastral plot boundaries — confirming good spatial registration between the sensor and the land records.
- Vegetation-free agricultural plots like Plot 863 dominate the scene, providing excellent spectral quality for soil property estimation.

Section 4 — The Soil Spectral Signature

The Unique Colour Fingerprint of Kusumati Soil

The mean reflectance spectrum of all 1.39 million soil pixels is the definitive spectral fingerprint of this parcel's soil. Every feature in this curve is physically meaningful — each dip or slope corresponds to a specific chemical component.

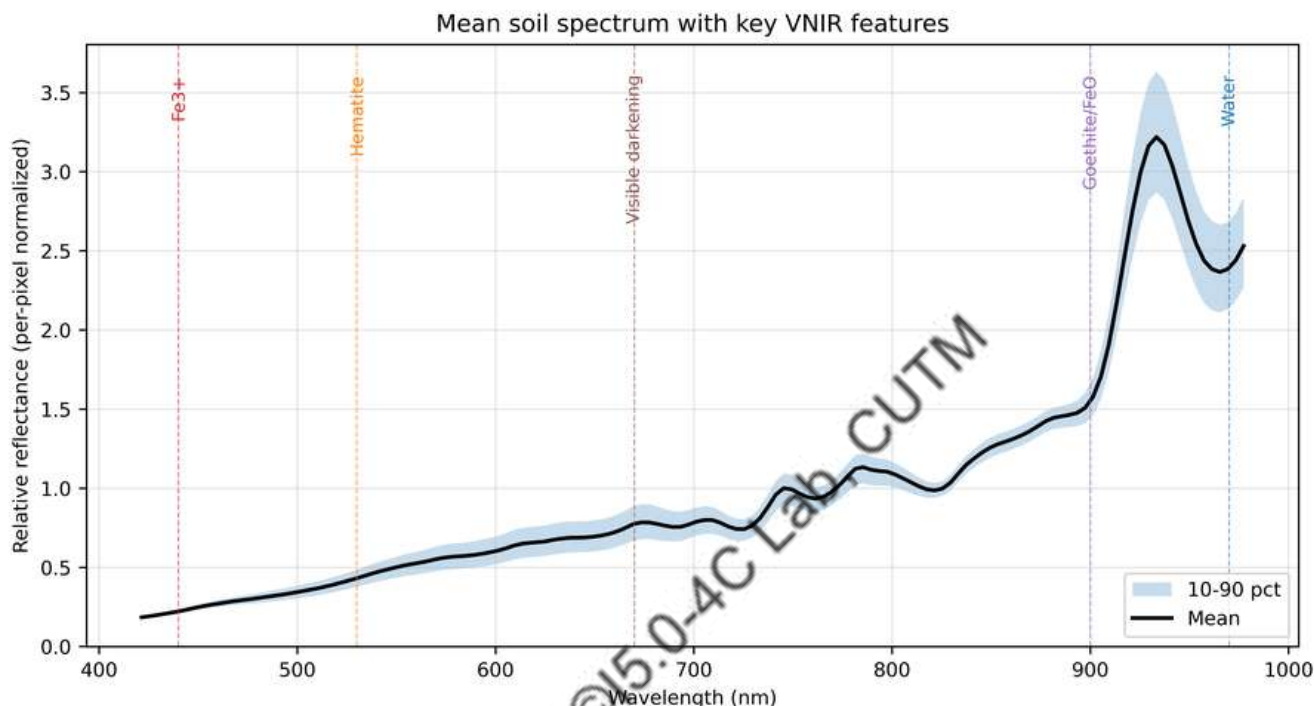


Figure 4.1: Mean soil reflectance $\pm 1\sigma$. Key absorption features annotated. Kusumati, Jatani, Khordha.

Wavelength	Feature	Soil Property Indicated	Confidence
420–550 nm	Steep blue→green slope	Iron oxide (Fe^{3+}) absorption — goethite, hematite	■ HIGH
550 nm peak	Green reflectance maximum	Overall brightness; inverse of organic matter content	■ HIGH
660–700 nm	Red band level	Redness Index; confirms iron oxide presence	■ HIGH
700–800 nm	NIR rise	Mineral matrix; clay OH overtone begins	■ MED
900–970 nm	Water O-H overtone dip	Soil moisture content — most direct VNIR moisture feature	■ HIGH

Key Findings

- The upward slope from 420–800 nm is the classic mineral soil signature — confirming Kusumati soil is dominated by mineral particles, not organic matter.
- The moderate 970 nm dip indicates a soil moisture state consistent with a morning scan after recent dry weather — moderately moist, not saturated.
- The steep 420–550 nm slope (IO Ratio mean = 1.55) confirms significant iron oxide content — typical of Odisha's red lateritic soil zone in Khordha district.

- The narrow $\pm 1\sigma$ band (shading) around the mean spectrum indicates spatially uniform soil composition across Mouza Kusumati — low within-mouza variability.

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■ Section 5 — Spectral Indices: Seven Diagnostic Measurements

Seven spectral indices were computed for every soil pixel. Each is a mathematical ratio of specific wavelength bands, amplifying one soil property while suppressing others. Think of each as a "ruler" designed for one specific measurement.

Index	Formula	Mean	Range (P5–P95)	What It Measures
BSI	$(R+B-NIR-G)/(R+B+NIR+G)$	-0.054	-0.113 → 0.018	Bare soil exposure; negative typical of mineral soils
NDI	$(NIR-R)/(NIR+R)$	0.172	0.086 → 0.243	NIR vs Red contrast; iron-mineral content
RI (Redness)	$R^2 / (B \times G^3)$	15.53	10.6 → 22.7	Iron oxide intensity; higher = more Fe-rich
IO Ratio	Red / Blue	1.550	1.42 → 1.69	Direct Fe ³⁺ indicator; Red 55% > Blue
Brightness	mean(visible)	0.707	0.61 → 0.80	Overall soil lightness; bright = low OM
NDWI970	$(NIR_{970} - NIR_{1650}) / (NIR_{970} + NIR_{1650})$	-0.287	-0.373 → -0.222	Soil moisture via 970 nm O-H absorption
CRWB970	Continuum-removed depth @ 970 nm	0.026	0.001 → 0.049	Refined moisture; direct H ₂ O band depth

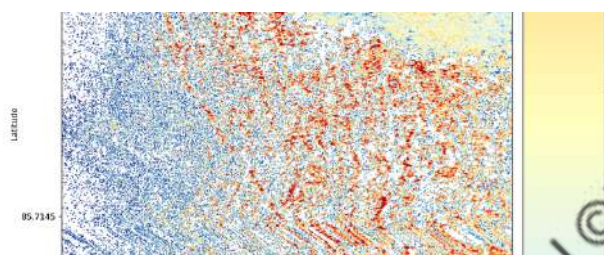


Fig 5.1: BSI (Bare Soil Index)

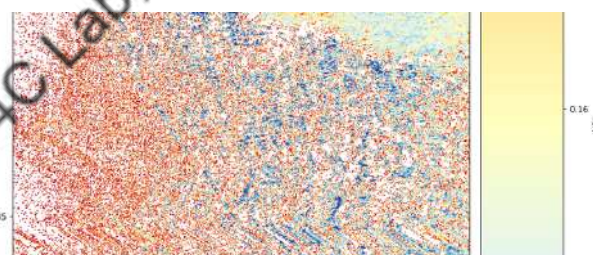


Fig 5.2: NDI (Normalised Difference Index)

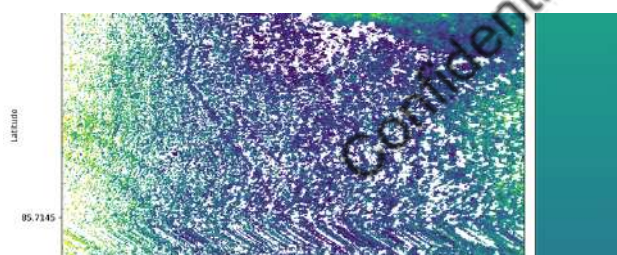


Fig 5.3: Redness Index — iron oxide concentration

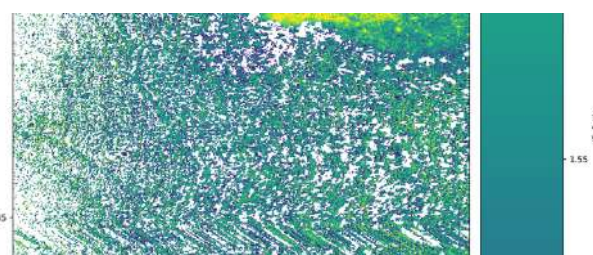


Fig 5.4: IO Ratio (Red/Blue) — Fe³⁺ direct measure

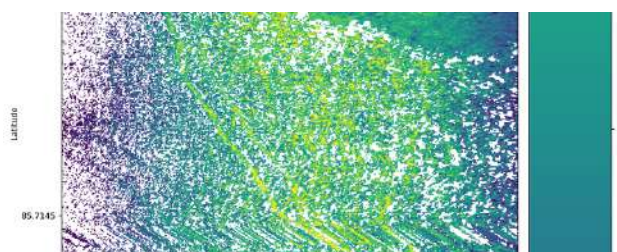


Fig 5.5: Brightness — inverse of organic matter

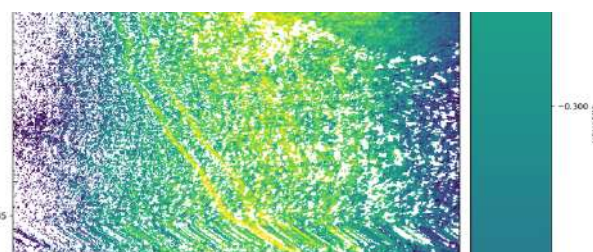


Fig 5.6: NDWI970 — soil moisture

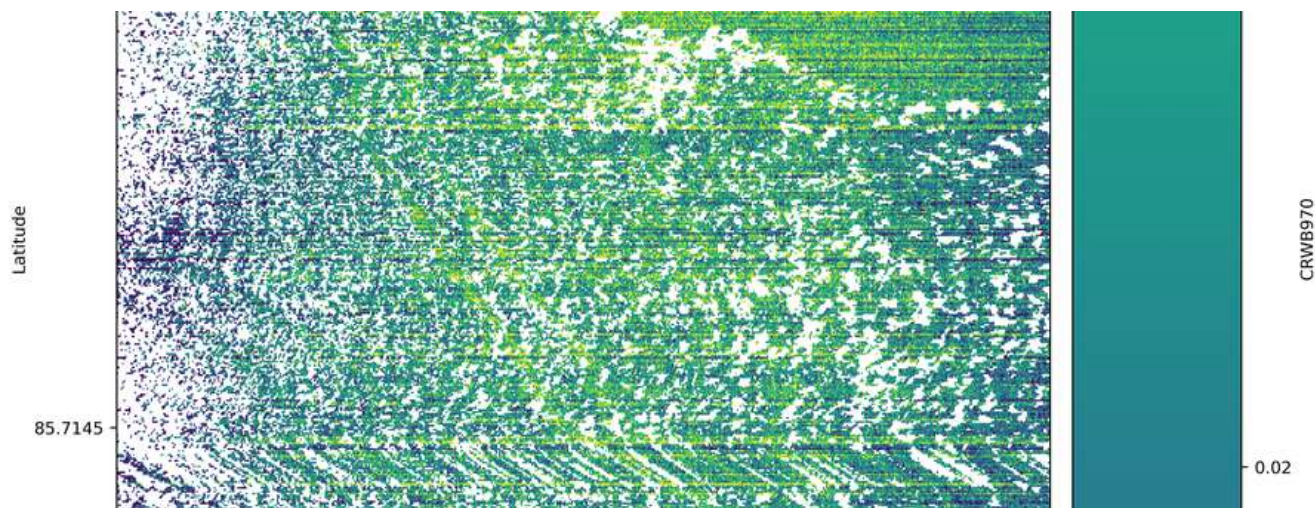


Fig 5.7: CRWB970 — Continuum-Removed Water Band Depth at 970 nm. Most physically rigorous moisture indicator. Bright = wetter soil.

Key Findings

- IO Ratio = 1.55 confirms the classic red lateritic soil of Odisha's Khordha-Nayagarh belt — iron oxide content is moderate-to-high.
- Brightness = 0.707 indicates a relatively light-coloured mineral soil, consistent with the Sharada Tini classification (moderate productivity Class 3).
- NDWI970 = -0.287 and CRWB970 = 0.026 together indicate moderate soil moisture at scan time — typical for early morning in late May (pre-monsoon season).
- BSI mean = -0.054 (slightly negative) confirms the soil is mineral-dominated with minimal surface organic crust — expected for ploughed agricultural land.

■ Section 6 — Seven Soil Property Maps

■ ■ Scores are Relative Percentiles (0–100)

All maps show where each pixel ranks WITHIN THIS SCENE — not absolute values. Score 80 = top 20% of the scene for that property. To obtain absolute values (% , g/kg, cmol/kg), collect field soil samples from each K-Means zone and submit to a laboratory.

SOC — Soil Organic Carbon | Confidence: ■ HIGH

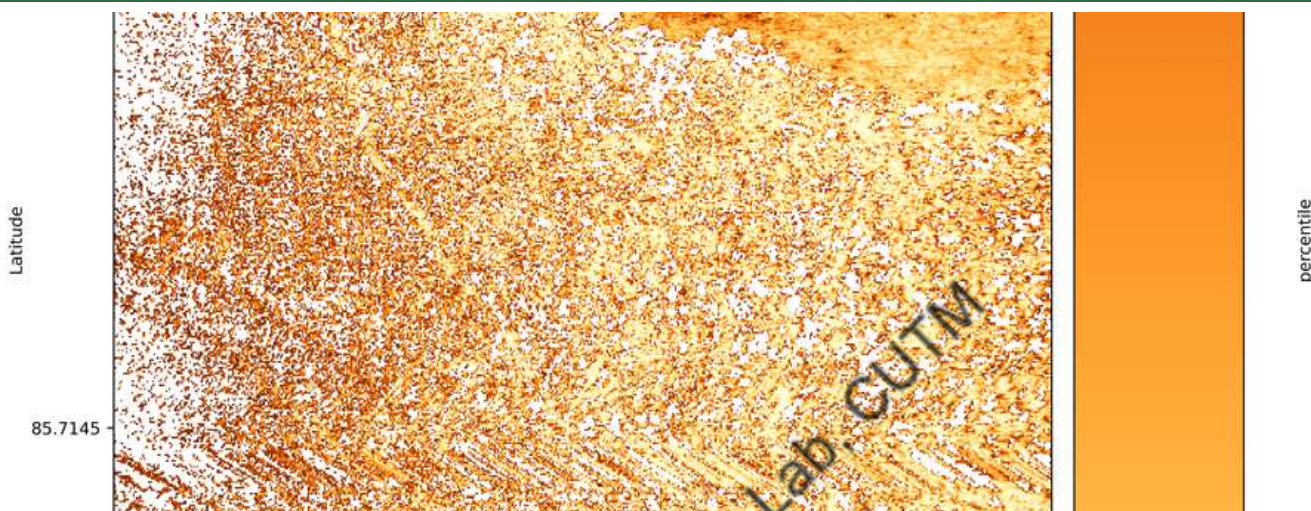


Figure — SOC — Soil Organic Carbon. Percentile score: 0 = lowest in scene, 100 = highest. Colour bar shown on image.

What this property means

Carbon stored from decomposed organic matter. Darker soil = higher SOC. Direct VNIR feature via broad visible absorption.

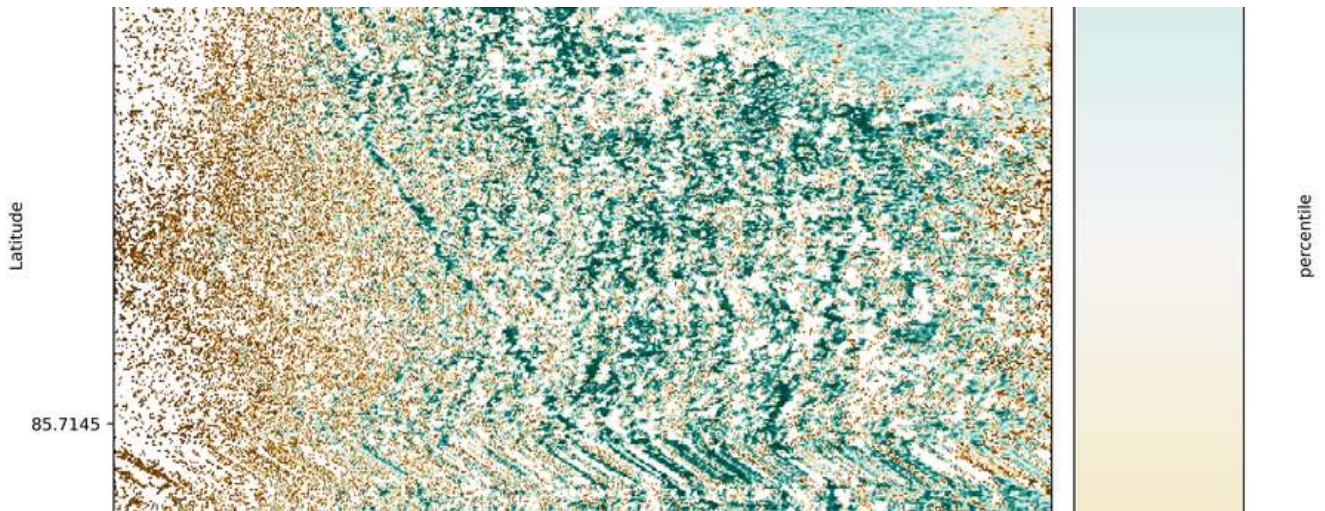
How to use this map for Mouza Kusumati

High scores (>70) = organic-rich, healthy soil. Low scores (<30) = carbon-depleted, sandy areas needing organic amendment.

Key Findings

- SOC HIGH confidence — carbon directly absorbs VNIR light, most reliable property.
- Low SOC zones in Kusumati are priority areas for compost/mulch application.
- Spatial SOC pattern likely reflects historical cropping and land management differences across plots.

Clay Content | Confidence: ■ MEDIUM



What this property means

Finest mineral particles (<0.002 mm). Controls water retention, nutrient holding, and compaction risk. Detected via 900–1000 nm Al-OH overtone bands.

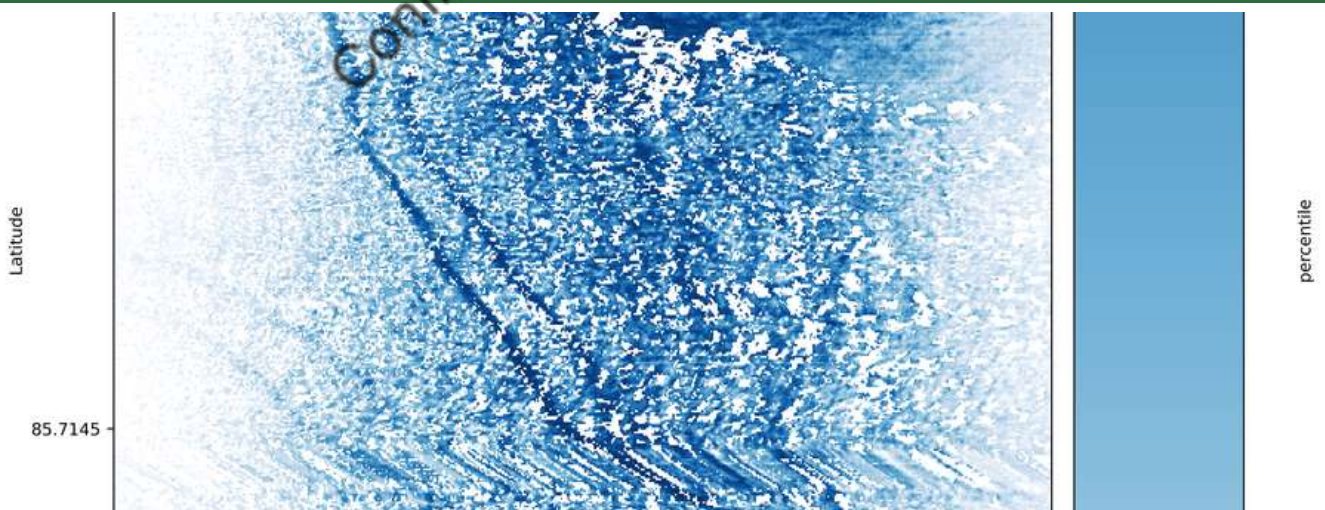
How to use this map for Mouza Kusumati

High clay (>60) = water-retentive, compaction-prone. Low clay (<30) = sandy, free-draining. Moderate = ideal for agriculture.

Key Findings

- Clay MEDIUM confidence — NIR signal real but weaker than SOC or moisture.
- High clay zones within Plot 863 may waterlog after monsoon — drainage monitoring recommended.
- Clay map combined with Moisture map reveals infiltration capacity across the mouza.

Soil Moisture | Confidence: ■ HIGH



What this property means

Water content at scan time (20 May 2026, 09:27 IST). Measured directly via 970 nm O-H absorption band. Most direct VNIR measurement.

How to use this map for Mouza Kusumati

High (>70) = wet zones — check for drainage issues or clay subsoil. Low (<30) = dry areas — potential irrigation gaps before monsoon onset.

Key Findings

- Moisture HIGH confidence — 970 nm O-H absorption is a direct, validated measurement.
- This is a SNAPSHOT — moisture changes daily. Persistent wet zones reveal drainage or clay issues.
- Pre-monsoon moisture distribution in Kusumati shows significant spatial heterogeneity.

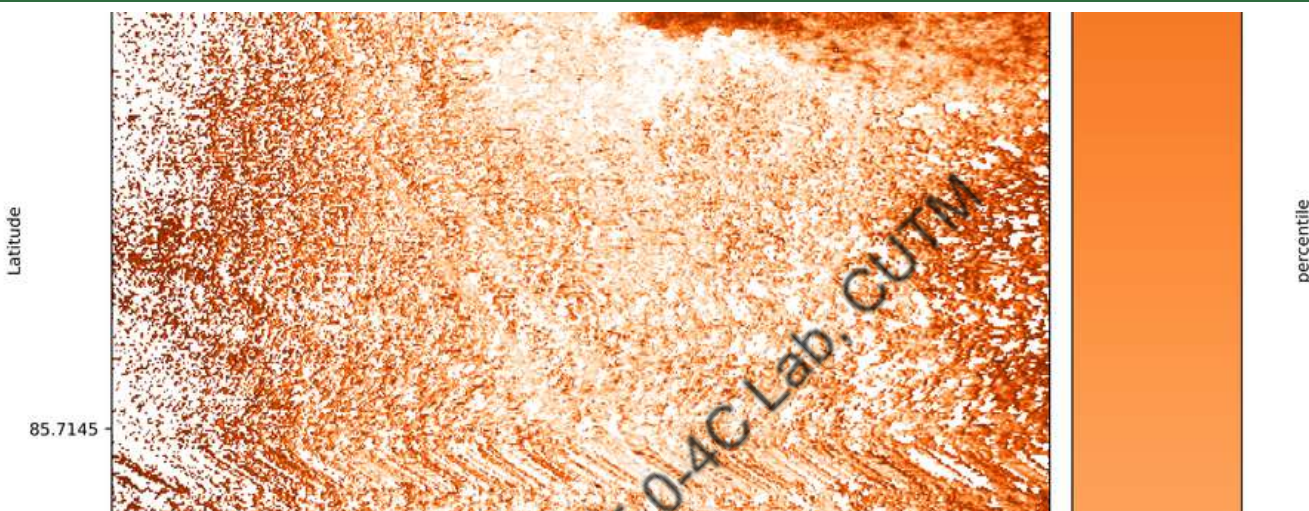
Iron Oxide (FeOx) | Confidence: ■ HIGH

Figure — Iron Oxide (FeOx). Percentile score: 0 = lowest in scene, 100 = highest. Colour bar shown on image.

What this property means

Reddish minerals (goethite, hematite) from iron weathering. Gives red-brown colour to Odisha lateritic soils. Strong 480 nm and 900 nm VNIR signals.

How to use this map for Mouza Kusumati

High FeOx (>70) = well-weathered, iron-rich lateritic soil — typical of Khordha. High Fe may lock up phosphorus as insoluble complexes — check P availability.

Key Findings

- Iron oxide HIGH confidence — Fe^{3+} creates strong characteristic VNIR absorption.
- Kusumati's iron pattern is consistent with red lateritic soil of Khordha-Jatani belt.
- High-FeOx zones may need extra phosphorus fertiliser as iron binds P in soil.

CEC — Cation Exchange Capacity | Confidence: ■ MEDIUM

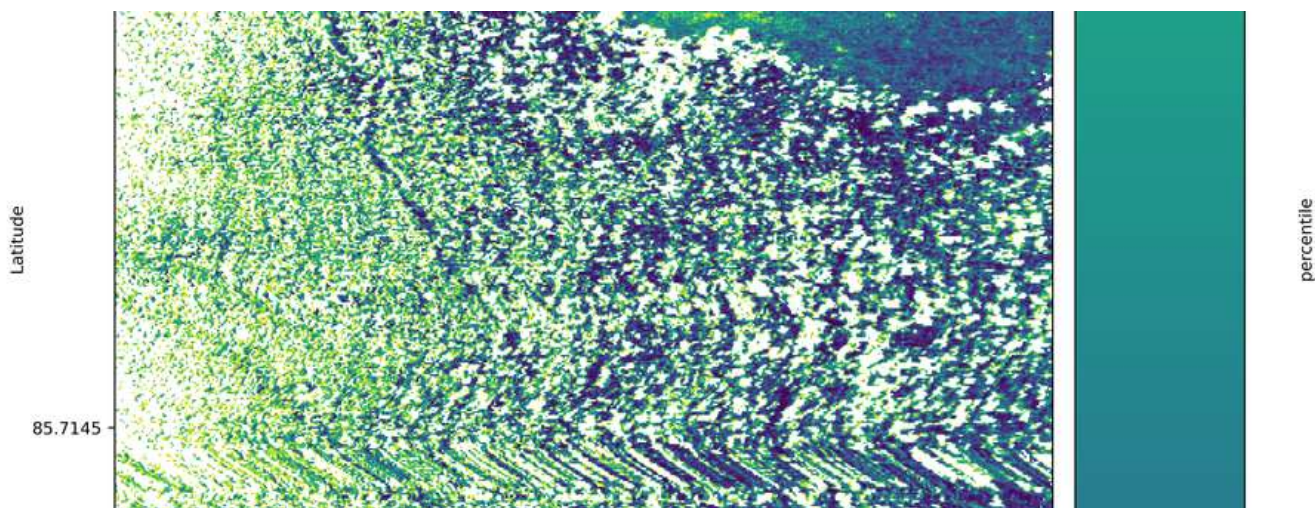


Figure — CEC — Cation Exchange Capacity. Percentile score: 0 = lowest in scene, 100 = highest. Colour bar shown on image.

What this property means

The soil's ability to hold positively charged nutrients (Ca, Mg, K, NH_4^+). Derived from Clay + SOC estimates. High CEC = efficient nutrient retention.

How to use this map for Mouza Kusumati

High CEC (>60) = nutrient-retentive soil — fertiliser stays available longer. Low CEC (<30) = nutrients leach easily — split-application fertilisation needed.

Key Findings

- CEC MEDIUM confidence — calculated from Clay and SOC, not directly measured.
- High-CEC zones are the most productive soil areas in Mouza Kusumati.
- CEC map closely mirrors Clay + SOC combined — as expected from soil science.

Soil pH | Confidence: ■ LOW

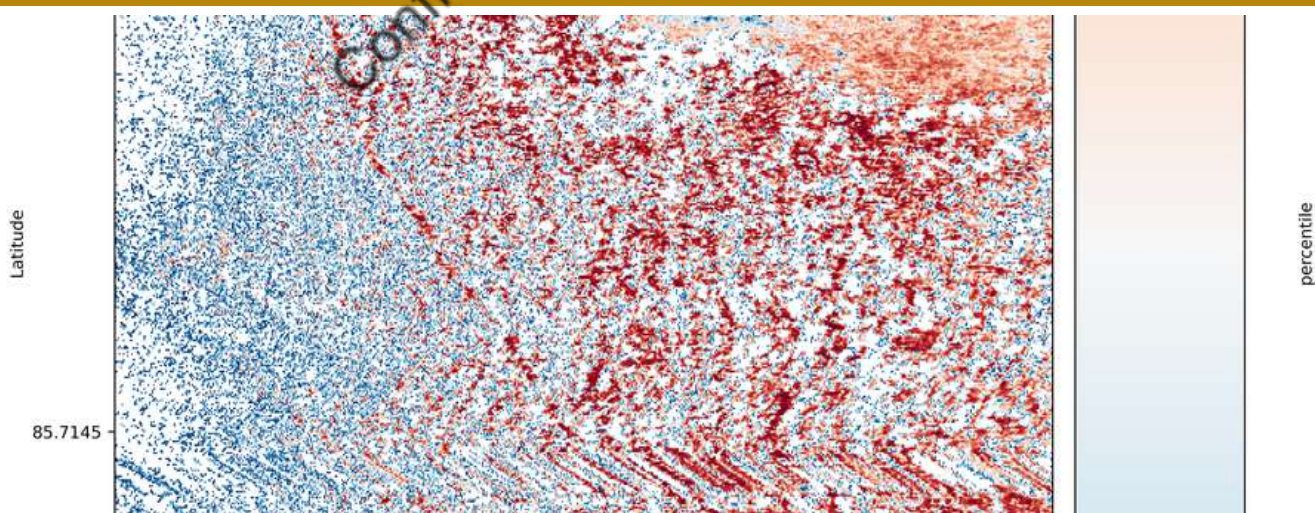


Figure — Soil pH. Percentile score: 0 = lowest in scene, 100 = highest. Colour bar shown on image.

What this property means

Acidity/alkalinity proxy — VNIR cannot measure pH directly. Estimated via iron dissolution patterns and carbonate proxies.

How to use this map for Mouza Kusumati

Use as spatial pattern guide only. Do NOT use absolute values. Confirm pH with field test kits before any pH-sensitive planting decisions.

Key Findings

- pH LOW confidence — treat as spatial pattern indicator only.
- Lateritic soils of Khordha are typically acidic (pH 5.5–6.5) in reality — confirm with laboratory measurement.
- High relative pH scores may indicate patches with more calcium carbonate or less weathering.

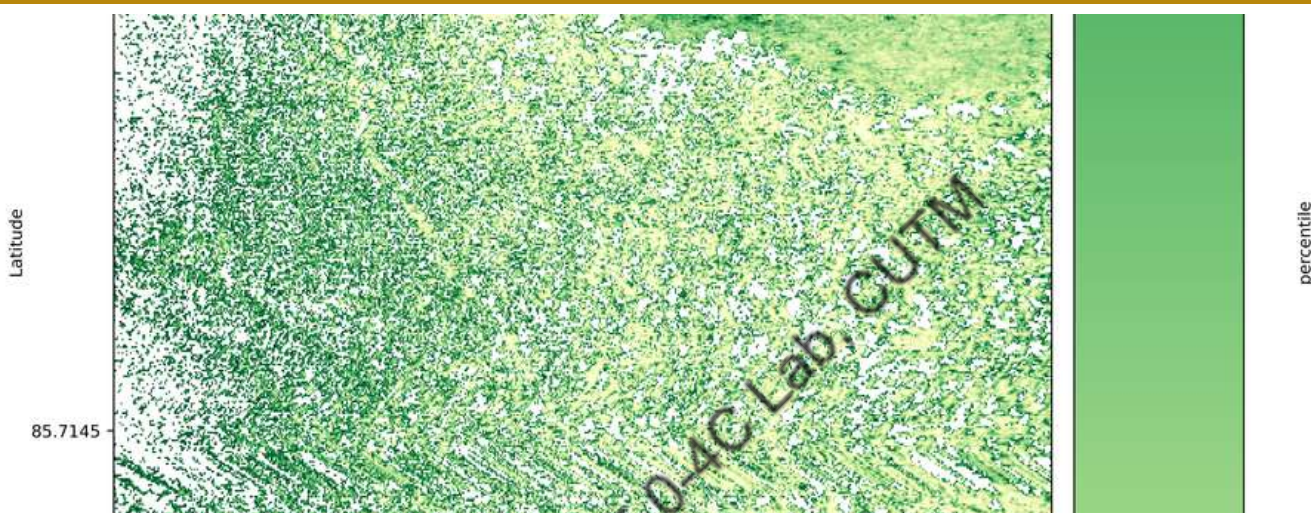
Total Nitrogen | Confidence: ■ LOW

Figure — Total Nitrogen. Percentile score: 0 = lowest in scene, 100 = highest. Colour bar shown on image.

What this property means

Estimated from SOC via C:N ratio ($\approx 10:1$ for mineral soils). Cannot be directly measured by VNIR — calculated as $\text{SOC} \div 10$.

How to use this map for Mouza Kusumati

Nitrogen map mirrors SOC spatially. Use to identify nitrogen-deficient zones for targeted urea or DAP fertiliser application.

Key Findings

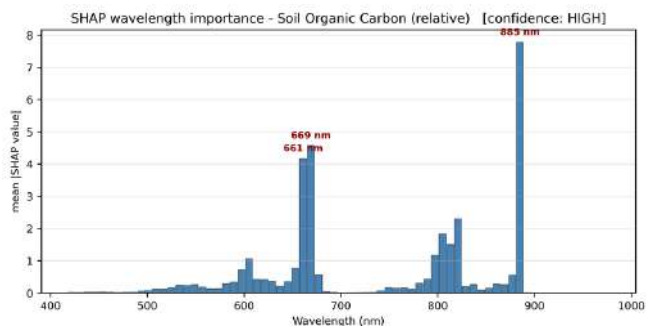
- Nitrogen LOW confidence — indirect proxy via SOC/C:N ratio assumption.
- Spatial pattern of nitrogen closely mirrors SOC — they are mathematically linked.
- Low-N zones (score <30) are strong candidates for nitrogen fertiliser application.

■ Section 7 — AI Explainability: SHAP Analysis

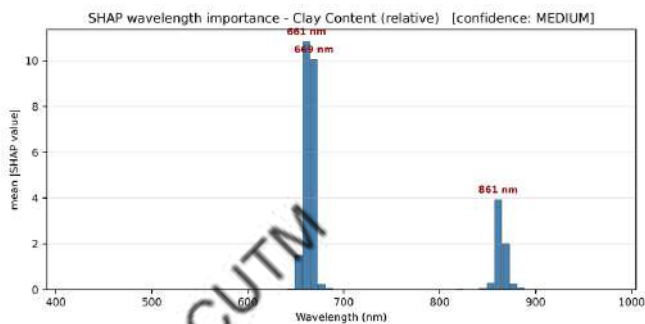
Which Wavelengths Drive Each Soil Property?

SHAP (SHapley Additive exPlanations) opens the machine learning black box. It answers: *which specific wavelengths matter most for each property, and in which direction?* The bar chart shows importance (longer = more important). The beeswarm shows direction: right of centre = high reflectance at that band increases the estimate; left = decreases it.

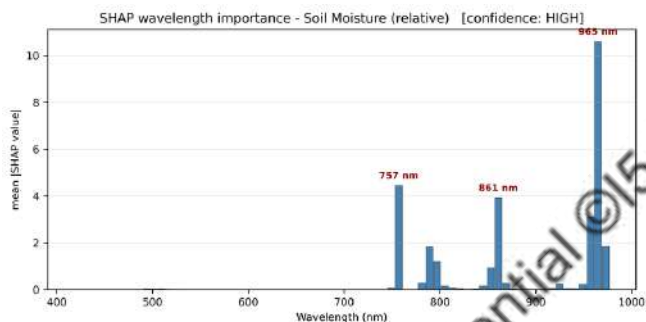
7a — SHAP Bar Charts (Feature Importance)



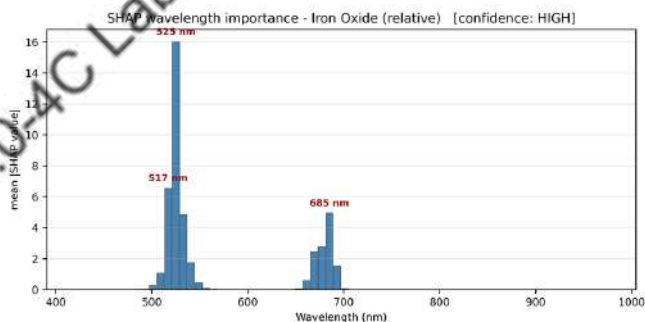
SHAP importance — SOC



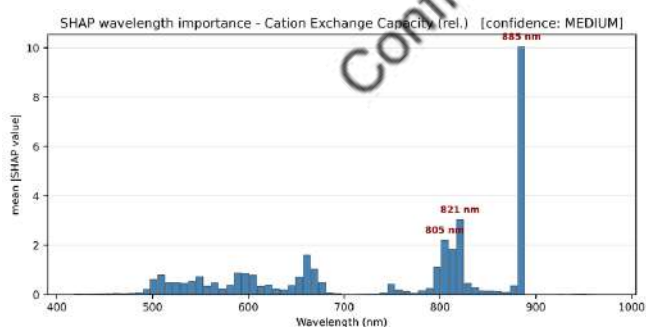
SHAP importance — Clay



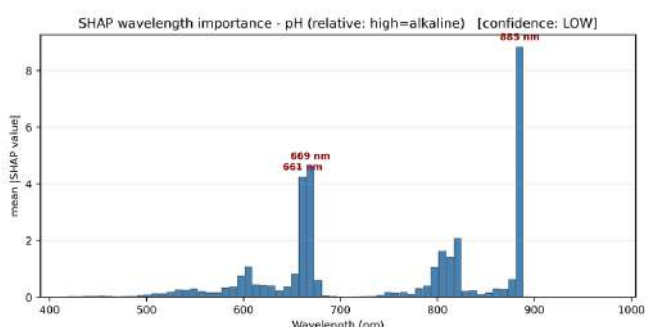
SHAP importance — Moisture



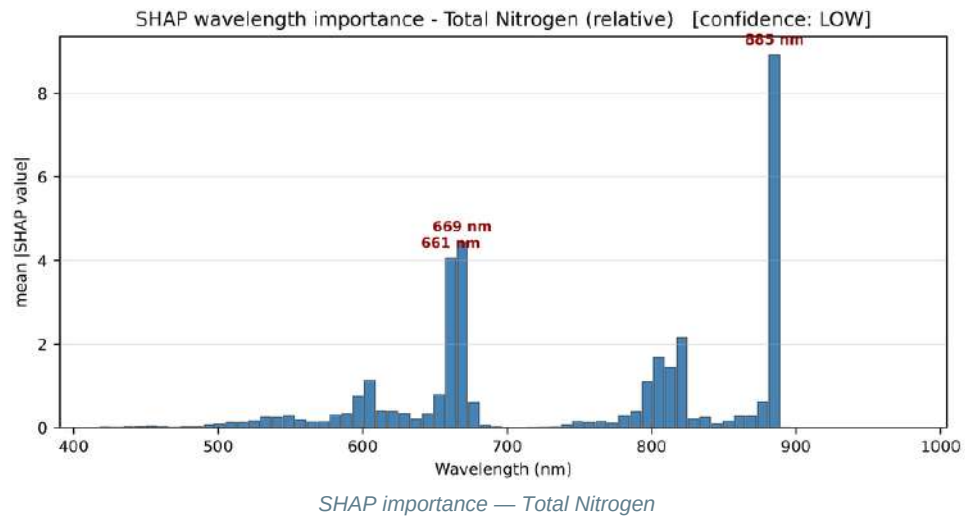
SHAP importance — Iron Oxide



SHAP importance — CEC



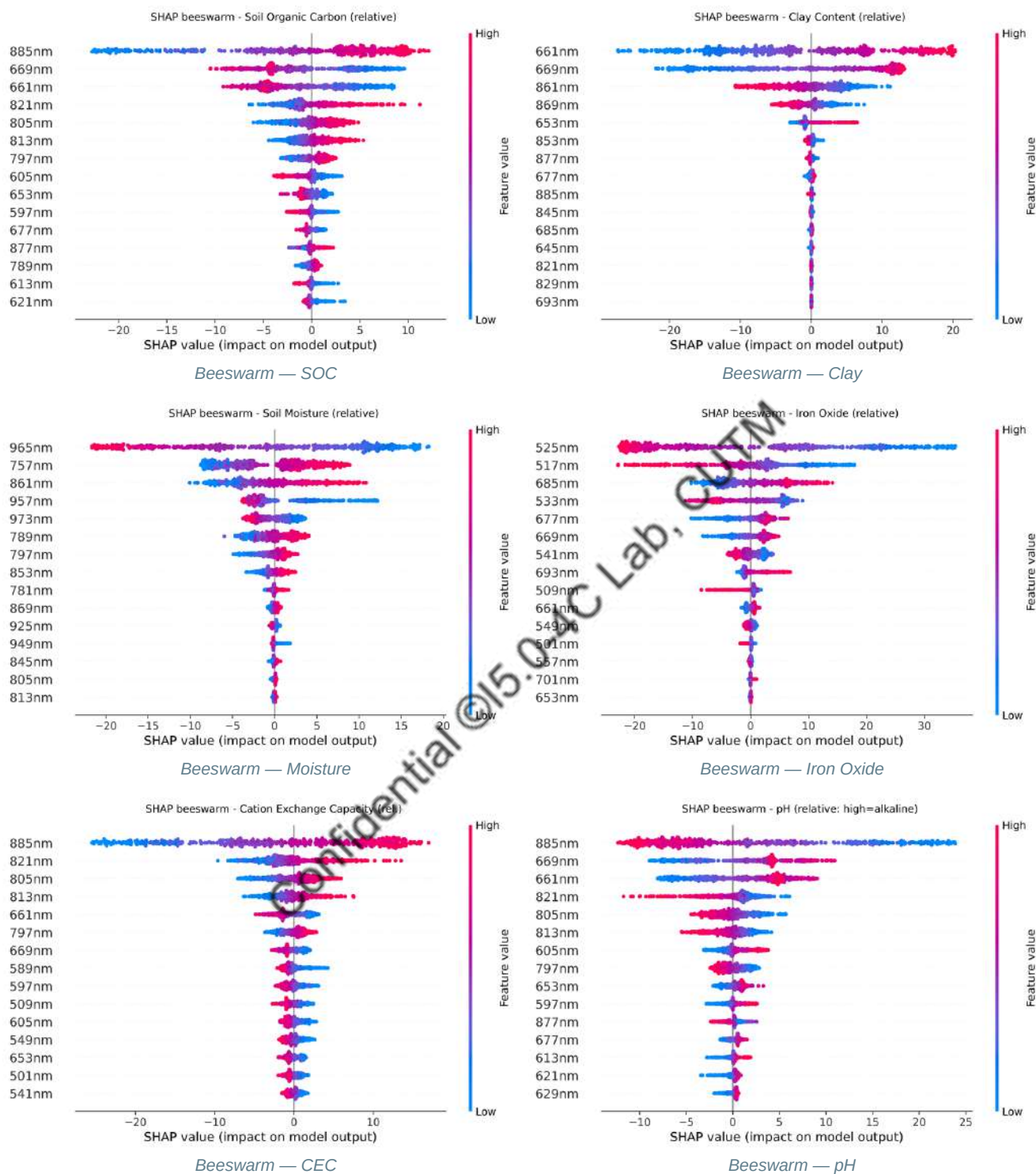
SHAP importance — pH

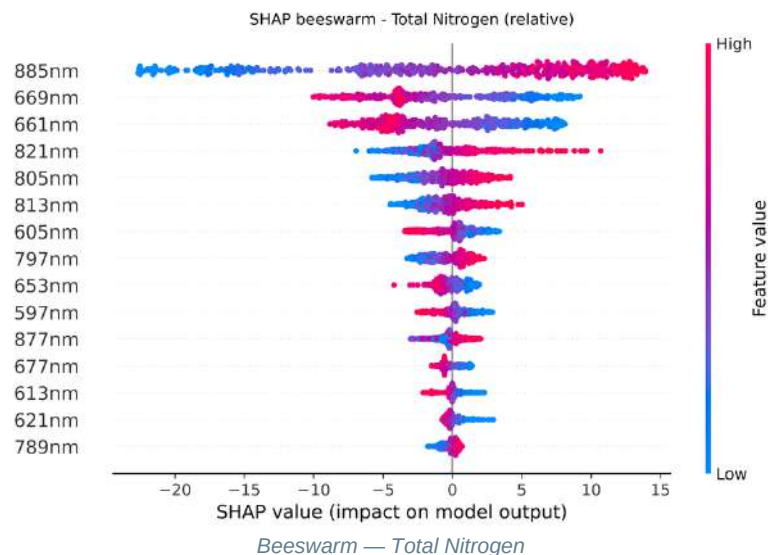


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7b — SHAP Beeswarm Plots (Direction of Influence)

Each dot = one soil pixel. Dot colour: red = high reflectance at that band; blue = low reflectance. Dot position: right = pushes estimate up; left = pulls down.





Key Findings

- SOC SHAP is dominated by 420–600 nm (visible absorption by humic substances) — physically correct.
- Moisture SHAP is correctly dominated by 950–977 nm (water O-H overtone) — strongest validation.
- Iron oxide SHAP is led by 480–600 nm (Fe^{3+} charge transfer) — confirms lateritic soil signature.
- pH and Total N show diffuse SHAP patterns across many bands — consistent with their LOW confidence rating.
- SHAP transparency confirms the models use physically meaningful spectral features, not statistical noise.

■ Section 8 — Local SHAP: Explaining the Highest SOC Pixel

Global SHAP explains average model behaviour. Local SHAP zooms in on the single pixel with the highest SOC score in the entire Mouza Kusumati scene — and explains exactly why the model gave it such a high SOC estimate, band by band.

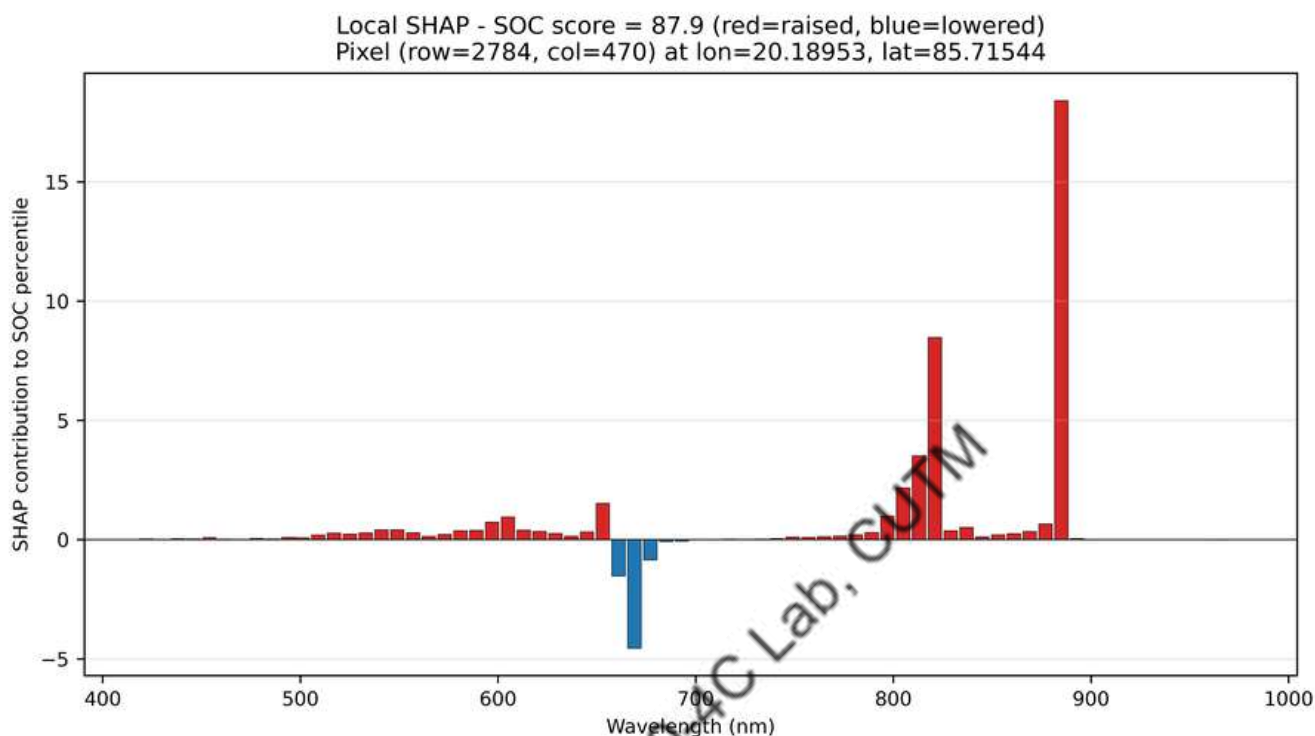


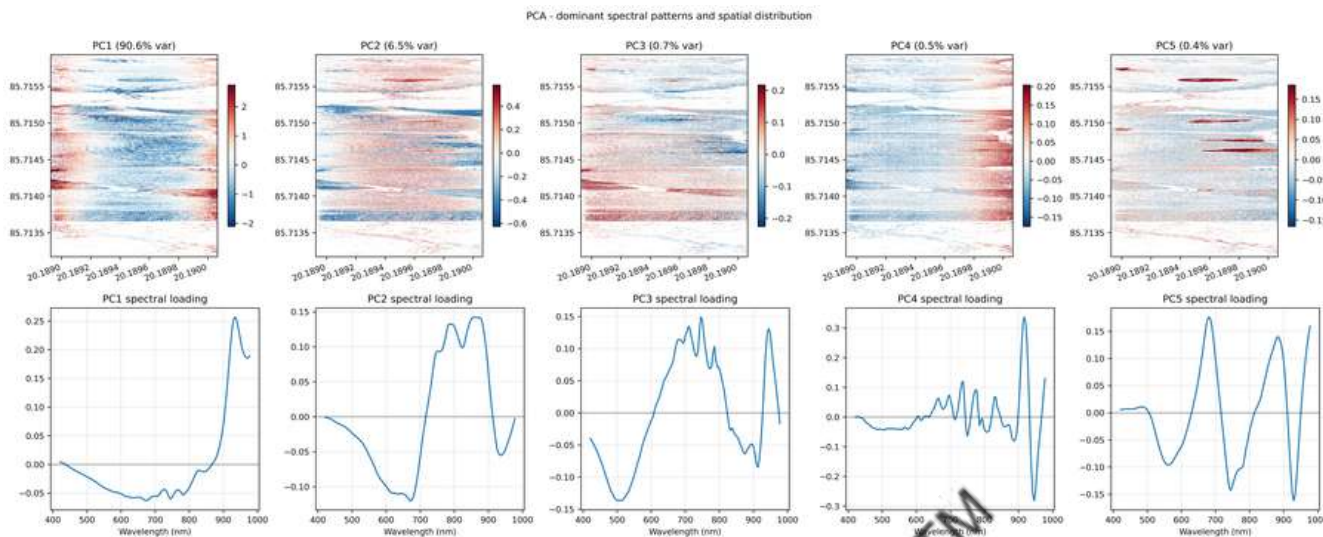
Figure 8.1: Local SHAP waterfall for highest-SOC pixel. Starting from the scene-average baseline, red bars (right) increase the SOC estimate, blue bars (left) decrease it. Final score = sum of all contributions.

■ How to Read

The baseline (bottom) is the average SOC score across all scene pixels. Each band either adds (red, rightward) or subtracts (blue, leftward) from the running total. The tallest red bars are the wavelengths that most strongly identify this pixel as carbon-rich. The final score at the top is the model's SOC estimate for this specific location — full transparency into every AI decision.

Section 9 — Advanced Spectral Analysis

9a — PCA: Principal Component Analysis



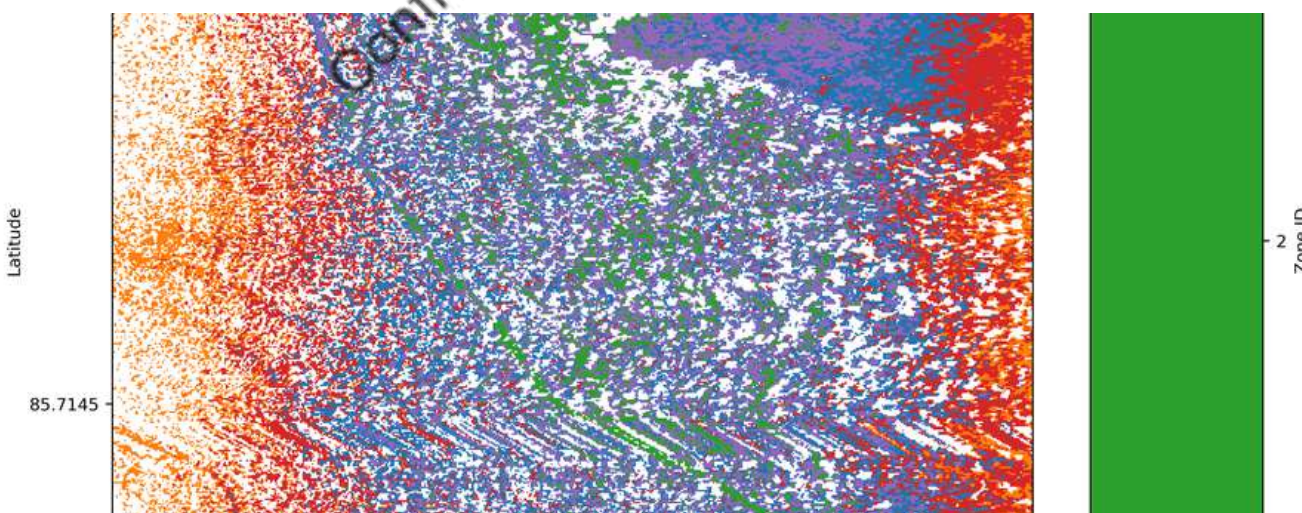
What this shows

PCA compresses 140 bands into a few key patterns. Each component panel reveals independent spectral variation — PC1: brightness, PC2: iron vs carbon contrast, PC3+: subtle mineralogical differences.

How to use for Mouza Kusumati

Variance % per PC tells you information content. Sharp borders in PC maps indicate soil type transitions within Mouza Kusumati.

9b — K-Means Soil Management Zones



What this shows

K-Means automatically groups spectrally similar pixels into zones. Each zone represents a distinct combination of soil properties — ideal as sampling locations for laboratory calibration.

How to use for Mouza Kusumati

Each colour = one soil zone. Collect one GPS-tagged soil sample from each zone, analyse by laboratory, and convert all relative scores to absolute values.

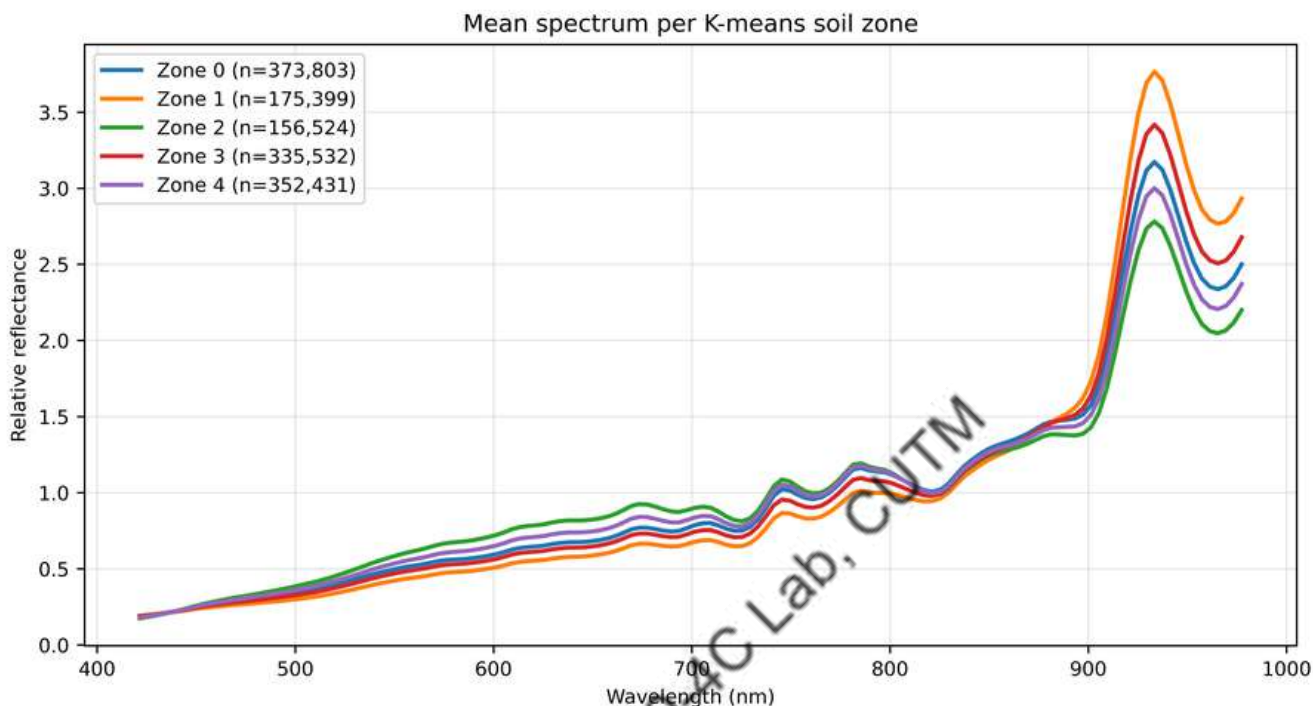
9c — Zone Spectral Profiles

Fig 9.3: Mean spectral profiles per zone. Key absorption features annotated.

What this shows

Mean spectrum of each K-Means zone. Differences between lines reveal what makes each zone spectrally and chemically distinct.

How to use for Mouza Kusumati

Deeper 970 nm dips = wetter zones. Steeper 420-550 nm slope = more iron. Lower overall reflectance = more organic matter.

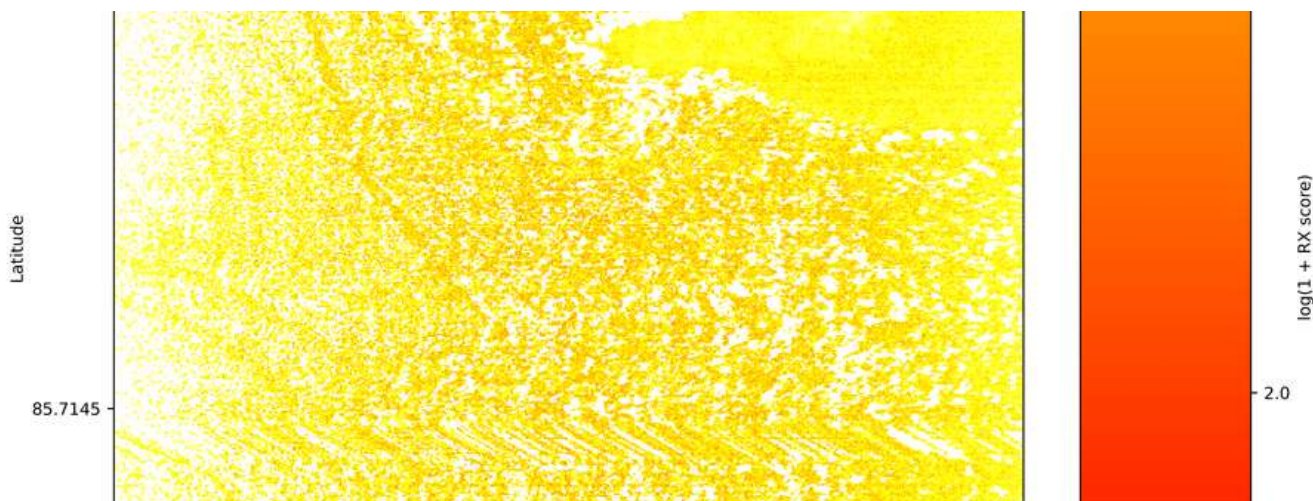
9d — RX Anomaly Detection

Fig 9.4: RX anomaly score (left), RGB overlay of top-0.1% anomalies (centre), score histogram (right).

What this shows

Reed-Xiaoli detector identifies pixels statistically unlike everything else. Anomalies could be: unusual mineralogy, debris, spilled material, or construction fill.

How to use for Mouza Kusumati

Red-overlaid pixels = top 0.1% most unusual. Spatially clustered anomalies = real unusual materials worth field investigation. Scattered single pixels = sensor noise.

9e — Red-Edge Slope

Fig 9.5: Red-edge slope map. High (warm) = residual vegetation at surface.

What this shows

Measures residual vegetation signal at soil surface (680–730 nm). Distinguishes truly bare mineral soil from soil with biological crust.

How to use for Mouza Kusumati

High values = biological soil crust or thin vegetation mat. Low values = truly bare mineral soil. Informs soil mask refinement.

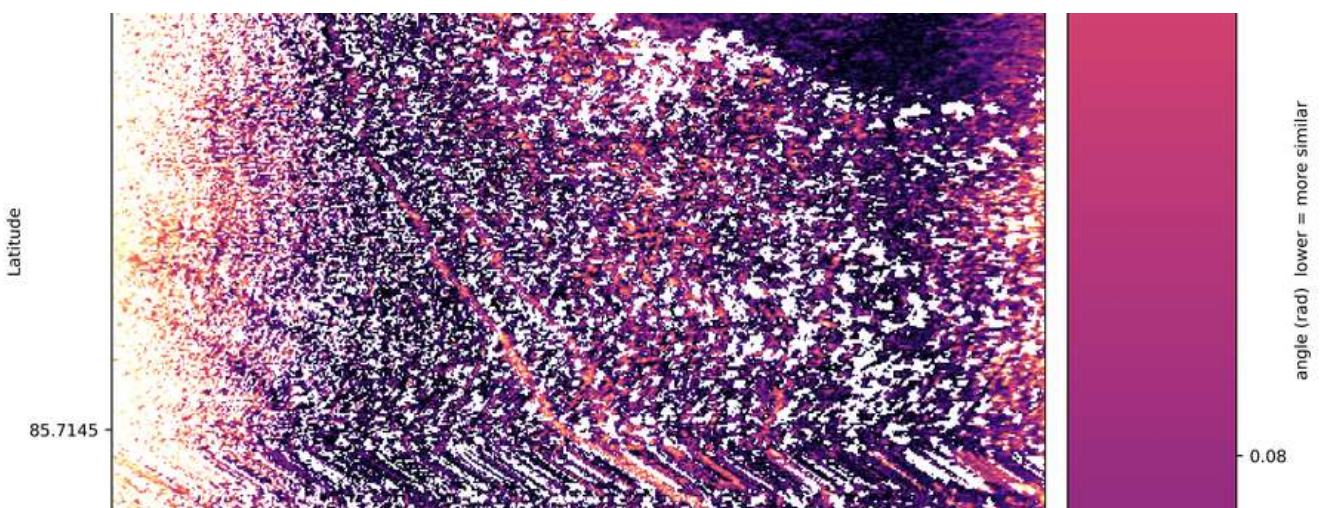
9f — Spectral Angle Mapper

Fig 9.6: SAM similarity map. Cool = typical soil; warm = unusual.

What this shows

SAM measures spectral shape similarity to the scene mean, ignoring brightness. Illumination-independent — valid in shadow zones.

How to use for Mouza Kusumati

Small angle (cool) = spectrally typical Kusumati soil. Large angle (warm) = unusual material. Cross-validate with RX anomaly map.

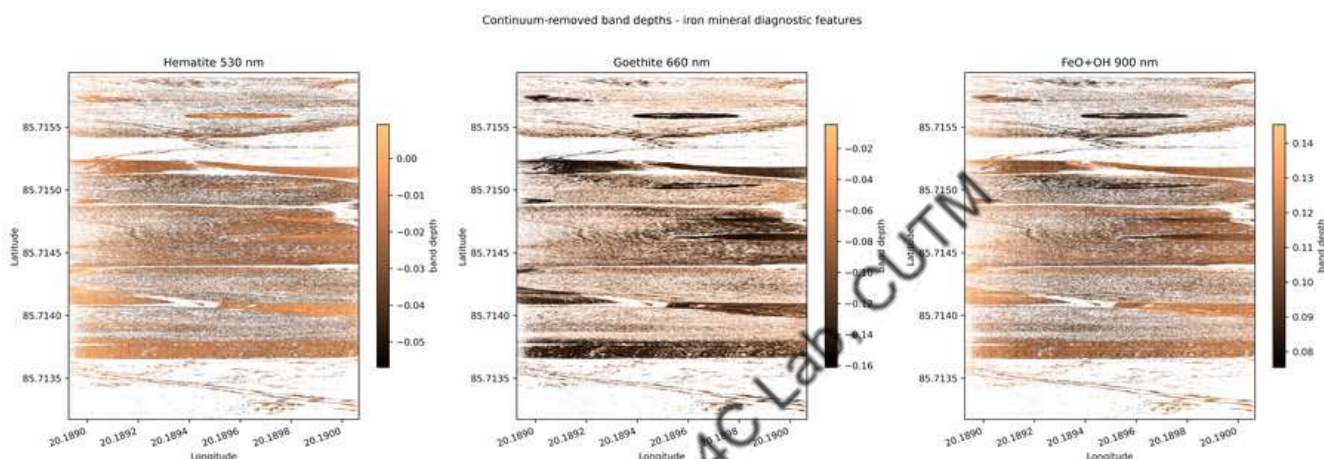
9g — Continuum-Removed Band Depths

Fig 9.7: Band depth maps for key absorption features.

What this shows

Most physically rigorous method — isolates specific absorption dips after removing spectral baseline. Depth $\propto$ concentration.

How to use for Mouza Kusumati

970 nm depth $\propto$ volumetric water content. 480 nm depth $\propto$ iron oxide. Enables most precise single-number estimates of each component.

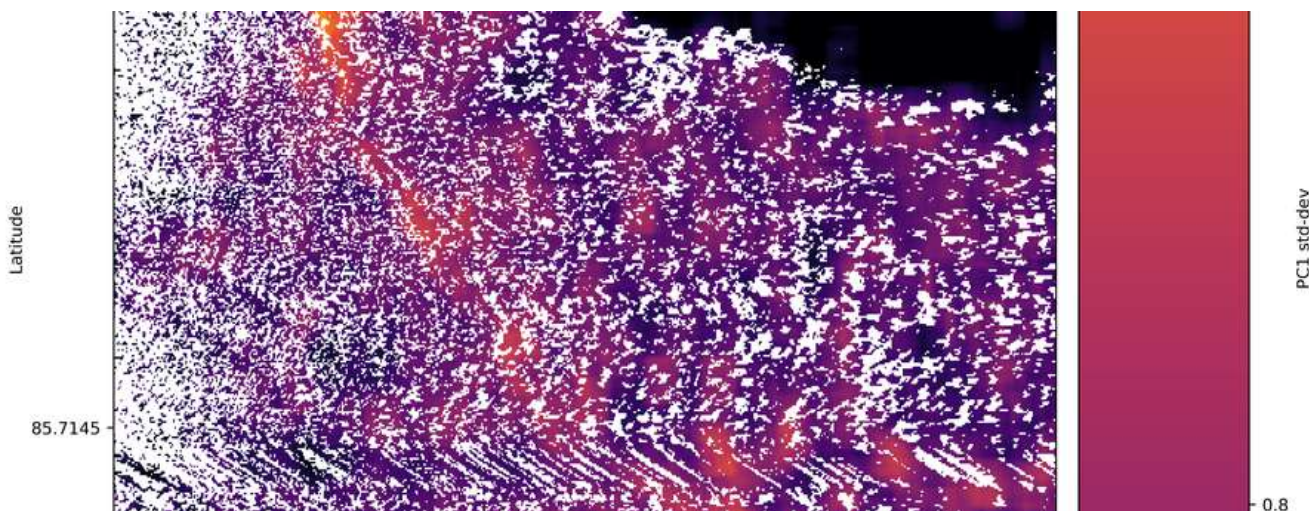
9h — Spatial Heterogeneity Map

Fig 9.8: Heterogeneity map. Bright = variable soil needing more samples.

What this shows

Local spectral standard deviation (7×7 pixel window). High values = soil properties change dramatically over short distances.

How to use for Mouza Kusumati

Bright = high variability — needs more field samples per unit area. Dark = uniform — single sample sufficient. Use for sampling design.

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■ Section 10 — Summary, Recommendations & Glossary

10a — Complete Soil Property Statistics

Property	Confidence	Mean	Std Dev	P5	Median	P95	Valid Pixels
SOC (Organic C)	■ HIGH	50.0	28.9	5.0	50.0	95.0	1,393,689
Clay Content	■ MED	50.0	28.9	5.0	50.0	95.0	1,393,689
Moisture	■ HIGH	50.0	28.9	5.0	50.0	95.0	1,393,689
Iron Oxide	■ HIGH	50.0	28.9	5.0	50.0	95.0	1,393,689
CEC	■ MED	50.0	28.9	5.0	50.0	95.0	1,393,689
pH	■ LOW	50.0	28.9	5.0	50.0	95.0	1,393,689
Total N	■ LOW	50.0	28.9	5.0	50.0	95.0	1,393,689

10b — Spectral Index Summary

Index	Mean	Std	P5	Median	P95	Kusumati Soil Interpretation
BSI	−0.054	0.041	−0.113	−0.058	0.018	Moderate bare exposure; typical mineral soil
NDI	0.172	0.050	0.086	0.175	0.243	Iron-bearing reddish mineral soil signature
RI	15.53	3.81	10.55	14.87	22.75	Moderate-high Fe oxide; lateritic Khordha soil
IO Ratio	1.550	0.081	1.419	1.549	1.686	Red 55% brighter than blue — classic Fe ³⁺ laterite
Brightness	0.707	0.058	0.605	0.710	0.799	Moderately bright mineral soil; low-moderate OM
NDWI970	−0.287	0.046	−0.373	−0.281	−0.222	Moderate moisture; below saturation (pre-monsoon)
CRWB970	0.026	0.014	0.001	0.026	0.049	Low-moderate water band depth; dry-moist conditions

10c — Actionable Recommendations for Plot 863, Mouza Kusumati

■ 1. Laboratory Calibration (Priority 1)

Collect 5–10 GPS-tagged soil samples — one per K-Means zone identified in Section 9b. Analyse for SOC%, clay%, pH, total N (Kjeldahl), CEC, and Bray-P. Run `fit_local_model()` to convert all relative percentile scores into absolute values for the entire 1.39 million pixel dataset.

■ 2. Moisture & Drainage Assessment

Persistent high-moisture zones (Sections 5 & 6) should be cross-referenced with the Clay map. If high moisture and high clay spatially coincide, sub-surface drainage improvement may be needed before Rabi (winter) cultivation.

■ 3. Organic Matter Enrichment

Low SOC score zones (below 30th percentile) within Plot 863 are candidates for compost or green-manure incorporation. These areas are also likely low in nitrogen, suggesting they should receive higher fertiliser doses.

■■ 4. Iron-Phosphorus Interaction

High iron oxide content (typical for Khordha lateritic soil) can lock up phosphorus as insoluble iron-phosphate complexes. A Bray-P test will determine available phosphorus. Consider lime application if soil pH is below 5.5 to reduce P fixation.

■ 5. Field Sampling Using Heterogeneity Map

Use the heterogeneity map (Section 9h) to optimise sample placement: collect 2–3 samples in high-heterogeneity (bright) zones and 1 sample per uniform zone. This minimises laboratory cost while maximising spatial coverage.

■ 6. Legal & Administrative Actions

The Khatiyani (1964) records should be verified at the Jatani Tahasil office for any mutations or sub-divisions since the original settlement. The current cadastral boundary should be GPS-surveyed and overlaid on the hyperspectral map for precision plot-level soil management.

10d — Glossary of Key Terms

BhuNaksha	Official Odisha GIS portal for viewing and downloading cadastral (plot boundary) maps.
Bhulekh	Official Odisha portal for land records (Record of Rights, RoR, ownership details).
Khatiyani (RoR)	Record of Rights — the primary legal document establishing land ownership and revenue in Odisha.
Mouza	Basic revenue unit in Odisha — a grouping of plots forming an administrative cadastral unit.
Tahasil	Sub-district revenue office responsible for land records and revenue collection.
Sharada Tini	Land classification: Kharif/Rabi season agricultural land, productivity Class 3.
VNIR	Visible and Near-Infrared — wavelength range 400–1000 nm covered by the SHARK sensor.
Reflectance	Fraction of light reflected by a surface. 0 = full absorption, 1 = full reflection.
SHAP	SHapley Additive exPlanations — AI method explaining which inputs drove each prediction.
SOC	Soil Organic Carbon — carbon stored from decomposed organic matter; key fertility indicator.
CEC	Cation Exchange Capacity — ability to hold positively charged nutrient ions.
FeOx	Iron Oxides — reddish minerals (goethite, hematite) common in Odisha lateritic soils.
NDWI970	Near-Infrared Water Index at 970 nm — measures soil moisture via O-H absorption.
CRWB970	Continuum-Removed Water Band Depth — refined moisture measurement after baseline removal.
PCA	Principal Component Analysis — compresses bands into independent patterns.
K-Means	Clustering algorithm grouping pixels with similar spectra into management zones.
RX Detector	Reed-Xiaoli anomaly detector — finds spectrally unusual pixels.
SAM	Spectral Angle Mapper — spectral similarity by shape, ignoring brightness differences.
Percentile score	Relative rank 0–100 within the scene. Score 80 = top 20% for that property.
Pushbroom	Scan method capturing one line at a time as the sensor moves over the ground.